

BILIM VA MALAKALARNI BAHOLASH AGENTLIGI DARAJASIDA

Bilimingga ishon va muvaffaqiyatga erish!

UMUMTA'LIM VA MAXSUS FANLARDAN BILISH DARAJASINI ANIQLASHNING MILLIY
SERTIFIKAT IMTIHONI ISHTIROKCHILARI UCHUN

MATEMATIKA
FANIDAN
TEST TOPSHIRIQLARI KITOBIDIR

Ishtirokchining familiyasi, ismi va otasining ismi

Imzo

Testni tuzuvchi: **TypoThesis jamoasi**

Telegram kanal: **@Milliymath_sertifikat**

Ushbu test varianti 45 ta (1-45) topshiriqdan iborat.

Test topshiriqlari RASCH MODEL asosida baholanadi.

Kitobning yopiq va ochiq turdagi test topshiriqlari mavjud.

– yopiq turdagi test topshiriqlarida javob variantlari (A, B, C va D) orasidan aynan bitta javobni tanlang va javoblar varaqasidagi topshiriq raqamiga mos qatorda belgilang.

– ochiq turdagi test topshiriqlarining javobini javoblar varaqasidagi topshiriq raqamiga mos qatorda aniq, tushunarli va eng sodda ko'inishda yozing.

– moslashtirishni talab qiluvchi yopiq test topshiriqlari uchun umumiy oltita (A – F) javob variantlari berilgan, uchta topshiriqqa (33-35 — test topshiriqlariga) ushbu javoblar orasidan mos ravishda bittadan javob tanlang va javoblar varaqasiga belgilang.

MILLIY SERTIFIKAT NAMUNAVIY SAVOLLAR TO'PLAMI!

7.03.2026. 1-smena.

1. $m = 24^3 \cdot 6^{12} \cdot 20^3$ bo'lsa, $\frac{m^2}{2^n}$ kasr qiymati natural son bo'ladigan n ning eng katta qiymatini toping.

A) 54
B) 60
C) 27
D) 48

2. $a = \lg 3$, $b = \cos 3$, $c = \ln 3$ sonlari uchun quyidagini soddalashtiring:

$$\frac{|a-b| + |b-c| - |a-c|}{|a+b| + |b-c| - |a+c|}.$$

A) $\frac{b+a}{b-a}$
B) $\frac{b-a}{c}$
C) $\frac{b-a}{a+b}$
D) $\frac{a-b}{a+b}$

3. Soddadil va mug'umbir bor edi. Mug'umbir soddadilni har ko'prikdan o'tganingda 2 marta pulingni ko'paytirib beraman dedi, evaziga 36 ming olaman dedi. va soddadil bunga ko'nadi. Soddadil ko'prikdan 3 marta o'tganda puli qolmabdi. Soddadilni puli qancha edi?

A) 31500
B) 31000
C) 32000
D) 30500

4. Tarkibida 60 l bo'lgan suv va neft aralashmasi tarkibidan 15 l neft ajratib olindi. Qolgan birikma tarkibida 4% neft qoldi. 60 l aralashmaning tarkibida necha foiz neft bo'lgan?

A) 20%
B) 24%
C) 26%
D) 28%

5. Soddalashtiring:

$$\frac{(a^{-2})^3 \cdot (-a^2)^7}{(-a^3)^2}.$$

A) a^2
B) a^3
C) $-a^2$
D) $-a^{-2}$

6. Hisoblang:

$$\frac{1}{\sqrt{2}+2} - \frac{1}{2-\sqrt{2}} - \sqrt{2}.$$

- A) $-\sqrt{2}$
- B) $-2\sqrt{2}$
- C) $2\sqrt{2}$
- D) $\sqrt{2}$

7. Hisoblang:

$$\frac{8(\sqrt{3}-\sqrt{15})}{\sqrt{3}+\sqrt{7}+\sqrt{11}+\sqrt{15}} + \frac{8(\sqrt{3}-\sqrt{15})}{\sqrt{3}-\sqrt{7}-\sqrt{11}+\sqrt{15}}.$$

- A) $9\sqrt{5} - 2\sqrt{77}$
- B) $9\sqrt{5} - \sqrt{77}$
- C) $-9\sqrt{5} + \sqrt{11}$
- D) $9\sqrt{5} + 3\sqrt{77}$

8. $\log_{12} 162$, $\log_{12} x$, $\log_{12} y$, $\log_{12} z$, $\log_{12} 1250$ sonlari arifmetik progressiyaning ketma-ket hadlari bo'lsa, $z - x - y$ ni toping.

- A) 30
- B) 36
- C) 32
- D) -30

9. $q < 1$ bo'lgan, geometrik progressiyaning hadlari uchun $b_1 + b_2 + b_3 = 21$ va $b_1 \cdot b_2 \cdot b_3 = 216$ bo'lsa, S_5 ni toping.

- A) 23
- B) 23, 25
- C) 23, 5
- D) 24

10. $a \geq 3$ uchun quyidagini soddalashtiring:

$$\left(\frac{a + \sqrt{a^2 - 3a}}{a - \sqrt{a^2 - 3a}} - \frac{a - \sqrt{a^2 - 3a}}{a + \sqrt{a^2 - 3a}} \right) \cdot \frac{3}{4}.$$

- A) $\sqrt{a^2 - 3}$
- B) $\sqrt{a - 3}$
- C) $a^2 - 3a$
- D) $\sqrt{a^2 - 3a}$

11. Soddalashtiring:

$$\frac{a^3 - b^3}{(2a + b)^2 - 3a(a + b)} + \frac{a^3 + b^3}{a^2 - ab + b^2}.$$

- A) $2a - 2b$

- B) $2b$
C) $2a + 2b$
D) $2a$

12. $a = \sin 17^\circ$, $b = \sin 167^\circ$, $c = \sin 375^\circ$ sonlarini kamayish tartibida yozing.

- A) $a > c > b$
B) $a > b > c$
C) $c > a > b$
D) $c > b > a$

13. Tenglamaning haqiqiy ildizlari yig'indisini toping:

$$\left(\sqrt{7+4\sqrt{3}}\right)^x + \left(\sqrt{7-4\sqrt{3}}\right)^x = 14 .$$

- A) 1
B) 2
C) 0
D) -4

14. Tengsizlikni yeching:

$$\sqrt{x-11} + \sqrt{12-x} > 0 .$$

- A) $[11; 12]$
B) $(11; 12)$
C) $(-\infty; 11) \cup (12; \infty)$
D) $\{11; 12\}$

15. Tengsizliklar sistemasi yechimlari $(x_1; y_1); (x_2; y_2); \dots; (x_n; y_n)$ bo'lsa, $x_1 + x_2 + \dots + x_n$ ni toping:

$$\begin{cases} x^2 + y^2 = 34 \\ x + xy + y = 7 . \end{cases}$$

- A) 6
B) -2
C) 0
D) 8

16. Tenglamaning haqiqiy ildizlari yig'indisini toping:

$$\left(\frac{2}{3}\right)^{\log_3(x+1)} \cdot \left(\frac{4}{9}\right)^{\log_{\frac{1}{9}}(x-1)} = \frac{2}{3} .$$

- A) 1
B) 2
C) 3
D) -1

17. Tengsizlikni yeching:

$$\begin{cases} \frac{36}{x} - x \leq 0 \\ \frac{|12 - 2x| \cdot (x^2 + 5x + 25)}{|x| - 3} \leq 0 \end{cases}$$

- A) $(-3; 0)$
- B) $(-3; 6)$
- C) $(-3; 0) \cup \{6\}$
- D) $(-3; 0) \cup (0; 6]$

18. Tenglamaning $[0; 4\pi]$ oralig'idagi ildizlari yig'indisini toping:

$$2(\sin^2 x + \operatorname{tg}^2 x) + 3 = 4\operatorname{tg} x + 2\sqrt{2} \sin x.$$

- A) $\frac{5\pi}{2}$
- B) $\frac{9\pi}{2}$
- C) 5π
- D) 9π

19. Tenglamaning haqiqiy ildizlari yig'indisini toping:

$$(x^2 - 8x + 12) \cdot \sqrt{10 + 3x - x^2} = 0.$$

- A) 5
- B) -2
- C) 3
- D) 11

20. $f(x) = x^2 - 2ax - a - 6$ funksiya Ox o'qini 2 ta nuqtada kesib o'tadi. Ox o'qini kesib o'tgan nuqtalar orasidagi masofa 10 bo'ladigan a ning qiymatlari yig'indisini toping. Bitta bo'lsa o'shani toping.

- A) -19
- B) 19
- C) -6
- D) -1

21. Agar $f(x) = \begin{cases} 8 + x, & x < 0 \\ 2 - x, & x \geq 0 \end{cases}$ bo'lsa, $f(f(f(-5)))$ ni toping.

- A) 9
- B) 3
- C) 7
- D) 5

22. $f(x) = \frac{x + x^2 + x^3 + \dots + x^{2024}}{x + x^2 + x^3 + \dots + x^{1012}}$ bo'lsa, $f'(1)$ ni toping.

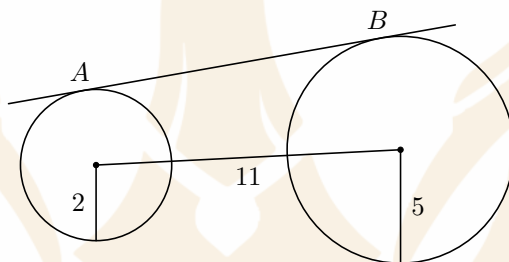
- A) 1012
- B) -1012
- C) 1
- D) -1

23. Hisoblang:

$$\int_3^6 \frac{x+1}{x^2+x-6} dx .$$

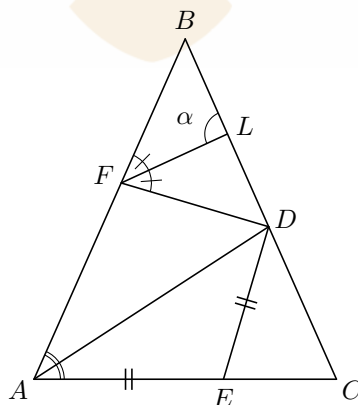
- A) $\frac{2}{5} \ln 6$
- B) $\frac{2}{5} \ln 12$
- C) $\frac{3}{5} \ln 12$
- D) $\frac{3}{5} \ln 6$

24. Quyidagi chizmaga asosan AB ni toping.



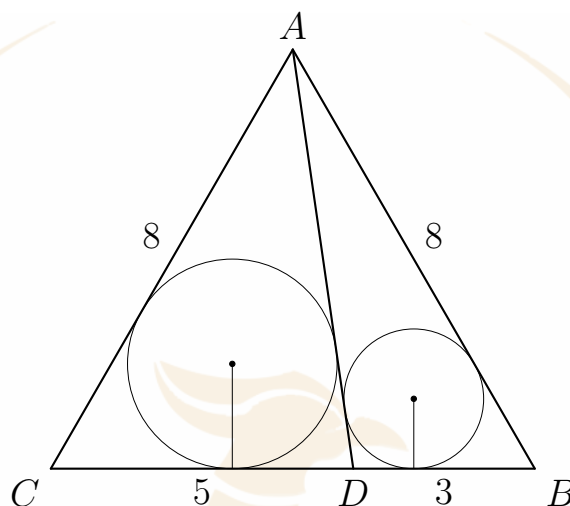
- A) $2\sqrt{7}$
- B) $4\sqrt{7}$
- C) $3\sqrt{7}$
- D) $5\sqrt{7}$

25. ABC uchburchak berilgan. BDC bir to'g'ri chiziqda yotadi va $|AC| = |AD|$, $|AE| = |DE|$ tengliklar o'rinli bo'lsa, $\angle FLB = \alpha$ ni toping.



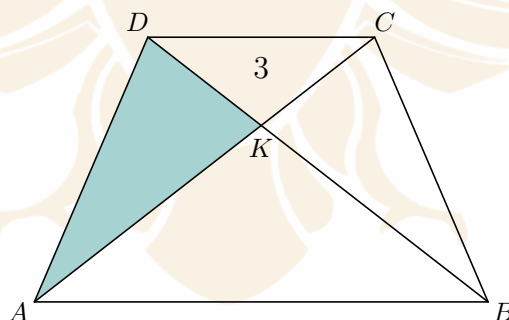
- A) 94°
- B) 105°
- C) 99°
- D) 72°

26. ABC teng tomonli uchburchakning tomoni 8 ga teng. AD kesma asosni 5 va 3 kesmalarga ajratadi. ADC va ABD uchburchakka ichki chizilgan aylana yuzalari yig'indisini toping.



- A) $\frac{14\pi}{3}$
- B) 4π
- C) 6π
- D) $\frac{13\pi}{3}$

27. $ABCD$ trapetsiyaning AC va DB diagonallari K nuqtada kesishadi. $S_{DCK} = 3$ va $S_{DCK} : S_{CKB} = 1 : 3$ bo'lsa, bo'yalgan soha yuzini toping.



- A) 7
- B) 8
- C) 9
- D) 10

28. Muntazam oltiburchakning tomoni 1 ga teng va uning tomonlari o'rtalari tutashirildi va yana o'rtalari tutashtirildi. Bu ish cheksiz davom etdi. Hosil bo'lgan yuzalar yig'indisini toping.
- A) $6\sqrt{3}$
B) $12\sqrt{3}$
C) $3\sqrt{3}$
D) $4\sqrt{3}$
29. $A(6; 4; -2)$, $B(x; y; 0)$ va $\overrightarrow{CD}(4; -3; 1)$ vektor berilgan. $\overrightarrow{AB}$ va $\overrightarrow{CD}$ vektorlar kollinear bo'lsa, $\overrightarrow{AB}$ ning koordinatalari yig'indisini toping.
- A) 12
B) 4
C) -12
D) -4
30. Tekislikda dioganallari 60 va 80 ga teng bo'lgan romb berilgan. Tekislikdagi tashqaridagi P nuqtadan dioganallari kesishgan nuqtagacha bo'lgan masofa 45 ga teng. P nuqtadan rombning tomonigacha bo'lgan eng qisqa masofani toping.
- A) 52
B) 54
C) 51
D) 53
31. $A = \{1, 2, 3, 4, 5, 6\}$ to'plam berilgan. To'plamning nechta qism to'plamida 3 yoki 5 dan kamida bittasi qatnashadi.
- A) 64
B) 48
C) 32
D) 56
32. $C_n^4 - C_n^3 = 14$ bo'lsa, C_n^2 ni toping.
- A) 24
B) 30
C) 32
D) 28

Topshiriqlar (33–35) va javob variant ($A–F$) larni o‘zaro moslashtiring.	Javoblar.
Balandligi 8 ga, to‘la sirti yuzi $232,5\pi$ bo‘lgan silindr berilgan.	A) $\frac{3\sqrt{21}}{4}$
33. Uchlari silindrning alohida asoslarida yotgan $AB = 10$ kesmadan silindr o‘qigacha bo‘lgan eng qisqa masofani toping.	B) $\frac{\sqrt{161}}{2}$
34. Silindr ichida kvadrat kesim olingan. Silindr o‘qidan shu kesimgacha eng yaqin masofani toping.	C) $\frac{19}{2}$
35. Silindrga tashqi chizilgan shar radiusini toping.	D) $\frac{3\sqrt{21}}{2}$
	E) $\frac{\sqrt{161}}{4}$
	F) $\frac{17}{2}$



36. Tenglamalar sistemasi berilgan:

$$\begin{cases} x^2 + y^2 - 2(x - y)a = 9 - 6a - a^2 \\ x^2 + y^2 + 2(3x + 4y)a = 1 - 2a - 24a^2 \end{cases}.$$

a) Tenglama bitta yechimga ega bo'ladigan a ning qiymatlari sonini toping.

Javob: _____

b) Tenglama bitta yechimga ega bo'ladigan a ning eng katta qiymatini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaraglasiga ko'chirib yozing.

37. Tenglamani yeching:

$$\sin^{10} x + \cos^{10} x = \frac{29}{16} \cos^4 2x.$$

a) Tenglamaning eng kichik musbat yechimini toping.

Javob: _____

b) Tenglamaning $[-\pi; \pi]$ oralig'idagi yechimlari sonini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaraglasiga ko'chirib yozing.

38. $f(x) = a|x+b|+c$ va $g(x) = kx+l$ funksiyalar $(6;1)$ nuqtada kesishadi. Funksiyalar $(-\infty;2)$ oralig'ida parallel bo'ladi va $f(x)$ funksiya $(2;3)$ nuqtada maksimumga erishadi.

a) $3 \cdot \frac{a+b+c}{k+l}$ ni toping.

Javob: _____

b) $(-\infty;2)$ oralig'ida $f(x)$ va $g(x)$ funksiyalar orasidagi masofani toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiiga ko'chirib yozing.

39. $f(x) = \frac{x^5}{40} + \frac{x^4}{8} - \frac{x^3}{6} - x^2 + 2$ funksiya berilgan.

a) Funksiyaning lokal maksimum nuqtalari sonini toping.

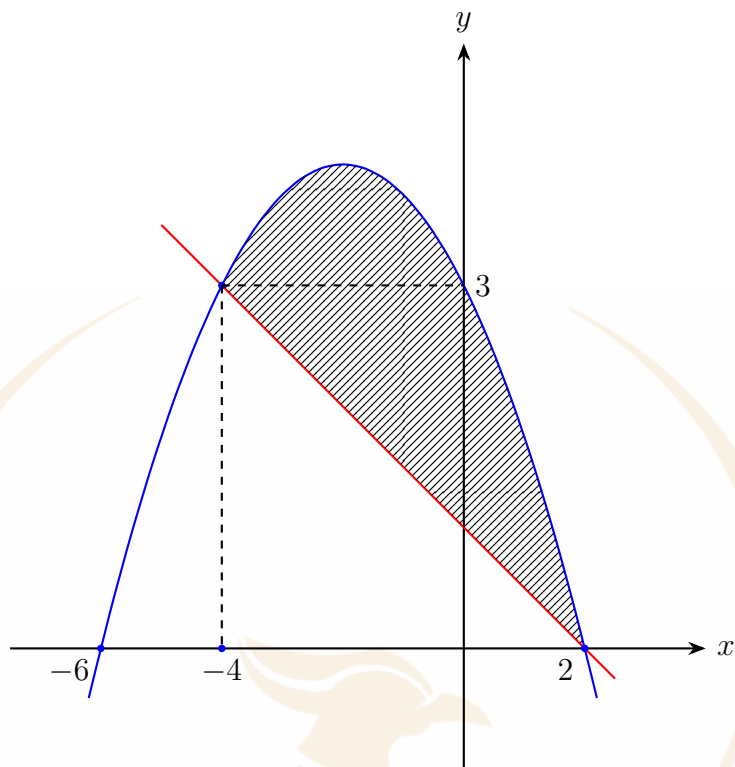
Javob: _____

b) Funksiyaning $[-5;2]$ orasidagi eng katta qiymatini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiiga ko'chirib yozing.

40. $y = ax^2 + bx + c$ parabola va $y = kx + l$ to'g'ri chiziqlar quyida tasvirlangan.



a) $\frac{a + b + c}{k + l}$ ni toping.

Javob: _____

b) Bo'yalgan soha yuzini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

41. To'g'ri burchak uchidan chiqan balandligi 6, bissektrisasi 8 bo'lgan to'g'ri burchakli uchburchak berilgan.

a) To'g'ri burchakli uchburchak yuzini toping.

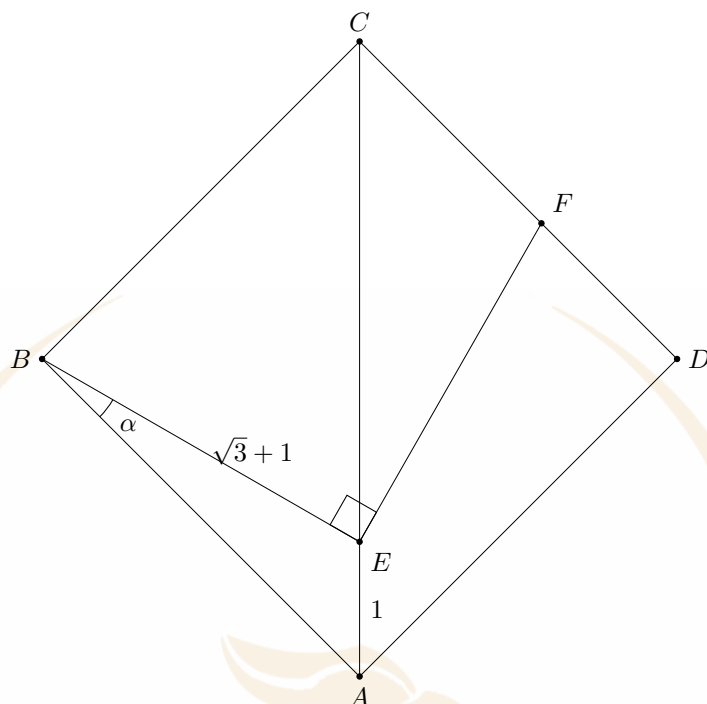
Javob: _____

b) Uchburchak gipotenuzasini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

42. $ABCD$ kvadratning AC diogonalidan E nuqta olingan. $BE = \sqrt{3} + 1$, $AE = 1$ va $BE \perp EF$ bo'lsin.



a) FD ni toping.

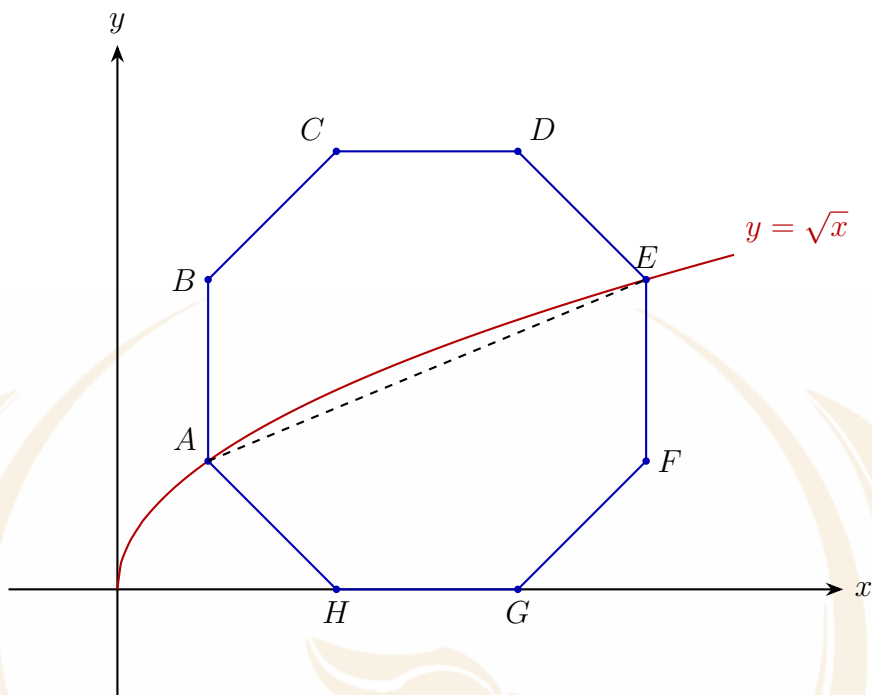
Javob: _____

b) $\angle ABE = \alpha$ ni toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

43. Quyidagi chizmada muntazam sakkizburchak va $y = \sqrt{x}$ funksiyalar tasvirlangan.



a) Muntazam sakkizburchak tomoni uzunligini toping.

Javob: _____

b) A va E nuqtalar orasidagi masofani toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

44. Muntazam to'rtburchakli piramidaga silindr ichki joylashtirilgan. Agar piramidaning asosini yuzi 144, hajmi 288 bo'lsa,

a) Silindrning eng katta hajmini qiymatini toping.

Javob: _____

b) Silindrning yon sirtini eng katta qiymatini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.



45. Anvar bankdan 60 million kreditni ikki yil muddatga yillik 20% bilan oldi. Uning oylik to'lovi quyidagi jadvalda ko'rsatilgan.

Oy	Umumiy qarz	Asosiy qarz	Foiz	Jami oylik to'lov
1	60 000 000	2 500 000	1 000 000	3 500 000
2	57 500 000	2 500 000	958 333,3(3)	3 458 333,3(3)
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				
21				
22				
23				
24				

a) O'ninchi oyda jami qancha pul to'laydi?

Javob: _____

b) Anvar bankga 24 oy ichida jami necha so'm foiz to'lagan.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasi ko'chirib yozing.